Manual

Connecting Kaba Termial for Time&Attendance collection

SCS-HCKP 8.1

Version 1.0.23049

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Copyright

Contents

1 Connecting the Kaba terminals to collect time & attendance

Purpose

You can connect hardware for personnel time recording (PZE) from Kaba with the product SCS-HCKP 3.0 to HYDRA.

Implementation notes

You use the product SCS-HCKP 3.0 if the following conditions apply:

- You want to connect the Kaba PZE hardware to HYDRA.
- You want to record personnel times (clock-in, clock-out, pause, and error reason clock-ins) on the Kaba PZE hardware and transfer the data to HYDRA for Personnel Time Management.
- You want to display account statuses managed in HYDRA on the KABA PZE hardware.

Integration

You need the KABA communication software *B-COMM* to use the product *SCS-HCKP 3.0*.

Features

- Interface for the transfer of clock-in, clock-out, break and error reason of the Kaba PZE hardware to HYDRA
- Interface for the transfer of account balances to the Kaba PZE hardware.
- Offline buffering of postings when HYDRA is not available.

2 KABA-Connector Collecting Working Hours (Time & Attendance)

The SCS-HCKP provides the license for the standard interface (KABA connector to collect working hours) between HYDRA and the B-COMM (communication software by dormaKABA). The KABA connector connects the PZE hardware from KABA to HYDRA to collect working hours. You can connect HYDRA with the KABA communication software B-COMM.

2.1 Requirements

- You must have a license for the communication software B-COMM (KB-BCOM25/KB-BCOME) and 3.16 newly must be installed in order to use the whole function range of the SCS-HCKP.
- If the B-COMM was installed by the customer, we advise to perform an analysis before the implementation.
- You also need the software option *B-Comm Parameter editor* for the B-COMM data communication. We also recommend to use the software option *B-COMM User management*. This option offers the possibility to protect the user interface of the *B-COMM (B-COMM GUI)* with a password.
- The software support the following badges:
    - MPDV Legic prime KGH with SSC 2C
    - MPDV "KGH" Legic Advant ISO 14443A with SSC 2C (for combi cards (Legic Prime/Advant)
    - MPDV "KGH" Legic Advant ISO 15693 with SSC 2C (for true Legic Advant badges)
- If you have terminals in different time zones, you need a HYDRA system for each time zone. The terminals, the KABA connector and the corresponding HYDRA system must operate in the same time zone.
- A combination between PZE and ZKS is not released for the same terminal.
- HYDRA service pack 13 must be installed and activated.
- You must update your terminals before the connection with the latest released program version from dormaKABA.

2.2 Supported terminal types

The following terminal types are released for SCS-HCKP:

Terminal
B-net 9340 HR2
B-web 9300 HR10
B-web 9600 K5 HR20
B-web 9700 K5 HR30

2.3 Administration

The KABA connector's installation is not included in the HYDRA installation. The connector is installed with B-COMM if required. The KABA connector is installed parallel to the B-COMM.

The KABA connector runs on the HYDRA server in Windows as a service or in Linux as a daemon. This service/daemon starts automatically when the server starts. The KABA connector operates independently, if HYDRA is started or not. It also works offline. If HYDRA is shut down, the KABA connector continues to accept postings (e.g. access logs) and stores this data on the hard drive. Once HYDRA is started, the connector transfers the postings to HYDRA.

2.4 Configurations in HYDRA

Most of the configuration collecting working hours (time & attendance) is executed in HYDRA. Only the language setting is configured in B-COMM. The following chapters describe the configurations and settings in HYDRA.

2.4.1 Authorization to clock-in and out

The assignment of clock-in and out authorizations for the personnel managed in HYDRA is performed Clocking authorizations in HYDRA via the application. You specify in the application Clocking authorizations the PZE terminal and where the person is permitted to clock-in.

2.4.2 Configuration of accounts

You make the general changes to the account names in the Configuration of accounts. In the application Configuration of accounts you can define the names of the individual accounts and, as a result, the names displayed on the KABA terminal.

2.4.3 Configuration of the account info

In the application Configuration of the account information you can specify the name of the individual accounts for a person, cost center, area, department, sub group, activity, employment status, or person does not clock-in. Therefore, you can Configuration of the account information specify the names for a subgroup on the KABA terminal. If there is no entry in the field, then the name is used.

2.4.4 Terminal configuration

2.4.4.1 Terminals

KABA terminal to collect working hours must be set up in the Terminal configuration . The following fields are relevant for these terminals:

Configurations in the tab *General*

Terminal

Terminal number to clearly identify

Active

Only active terminal are processed by the KABA connector.

Type

You store in this field the terminal type of the KABA terminal. The KABA connector only processes terminals that are assigned to the type KABA in the field *Type*.

Type	Description
80	KABA 9300
85	KABA 9600
90	KABA 9700

Terminal class

The terminal class specifies the terminal settings. Leave the terminal class in the standard empty so that the terminal class is filled with the value from the *Type* field. It may be necessary to store a different terminal class for customer-specific processing.

Operated as HYDRA-PZE/ZKS terminal

Activate the option *Operated as PZE/ZKS terminal.*

Cycle duration of status messages

Specification of the time interval in hours and minutes when the KABA connector reports the terminal status to HYDRA.

IP address

The terminal's IP address. For KABA terminals, enter the group ID and the DeviceID (GID and DID) of the terminal after the IP address, separated by a semicolon (example: 192.168.10.213; 0105).

Company number/system number

The value entered in this field can override the system number defined in the HYDRA Basic settings for individual terminals.

Configurations in the tab *HR functions*

Operation mode

Set the operating mode *Automatic status* or *Manual status changeover.*

Cyclic loading

This is the time when change in the authorization to clock-in or out is transferred to the terminal. If you check the option *Reload authorizations* in the *Terminal administration* of the *Terminal configuration*, you can define that modifications are transferred to the terminal at the latest after the end of the *Cycle duration of status messages* (if the terminal is online).

The KABA connector promptly reads changes in the configuration for terminals. In case of a **new** terminal, you must first define the terminal configuration and the access points. Then you must check the option *Reload program* in the *Terminal administration* of the *Terminal configuration* to transfer the configurations to the terminal.

Return time

You use this field to specify the time that the terminal waits before returning to default status (auto status, for example) after a key is pressed.

Terminal administration in the toolbar

Terminal from, to

You can change the *Terminal administration* for one or several terminals.

Activate terminal

Use the option *Activate terminal* to enable or disable in one action the field *Active* for all pre-selected terminals.

Reboot

If you check the option *Reboot*, all pre-selected terminals are rebooted. After the next status message, the connector informs the terminal about the reboot. The reboot is therefore executed within the time period entered in the field *Cycle duration of status messages* (if the terminal is online).

Next reboot on

In the field *Next reboot* you can enter the point in time of the reboot. Once this point in time is reached, the terminal is only rebooted after having sent the next terminal status.

Reload program

If you check this option, the complete HYDRA configuration is loaded on to the terminal. All badges on the terminal are first deleted and then reloaded. During the deletion process and the reboot the badges with authorization might not get access for several minutes. The access terminal is also rebooted which means the reader does not work for several minutes.

Reload authorizations

If you check the option *Reload authorizations*, all changes to accesses, access time models, opening hours and public holidays are transferred to the terminal. This at the latest after the end of the time entered in the field *Cycle duration of status messages* (see field *Cyclic loading* in the tab *HR functions* of the application *Terminal configuration*).

Authorizations are not located on the terminal, but the B-COMM. The option *Reload authorization* updates the clock-in and out authorizations in the B-COMM.

2.5 Configuration of the B-COMM

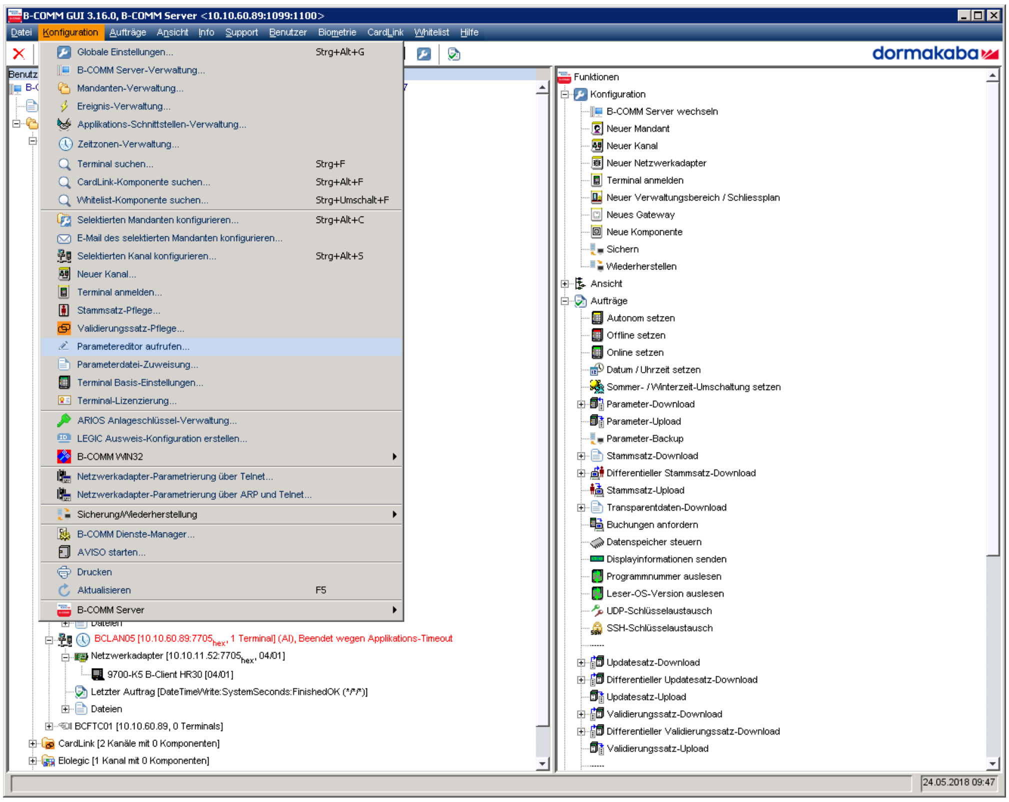
2.5.1.1 Language settings

The language setting of the KABA Terminal are performed in the B-COMM.

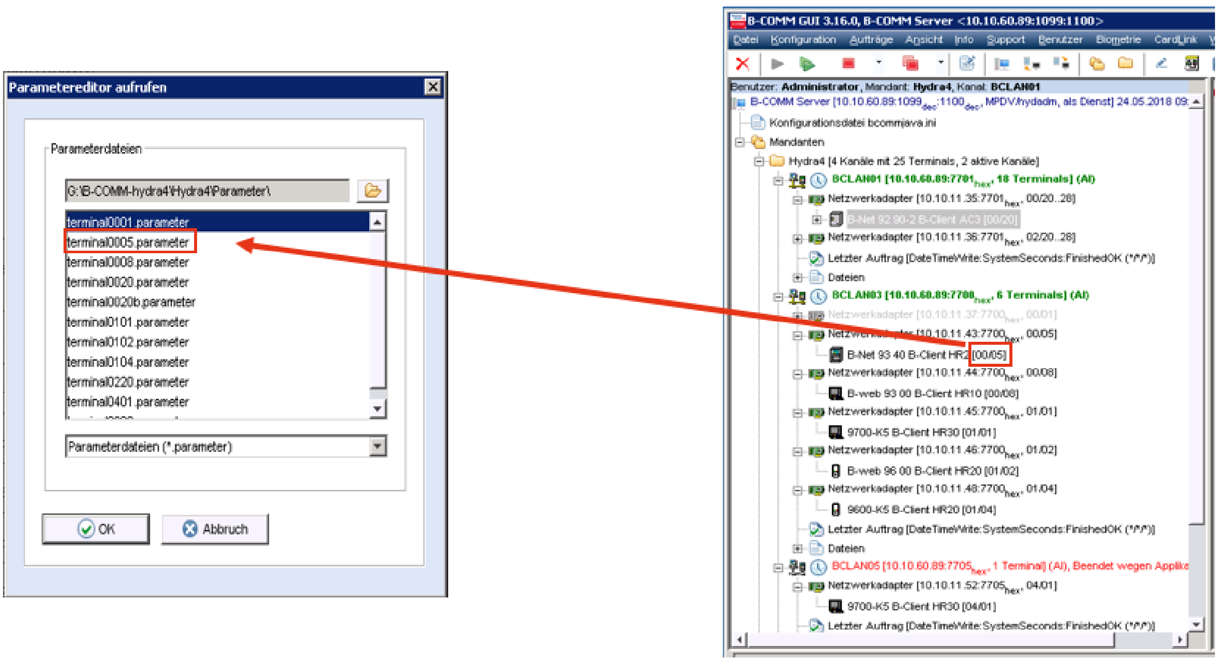
The standard setting is in German, English and French. You can switch the language on the KABA terminal.

Proceed as following to set the language:

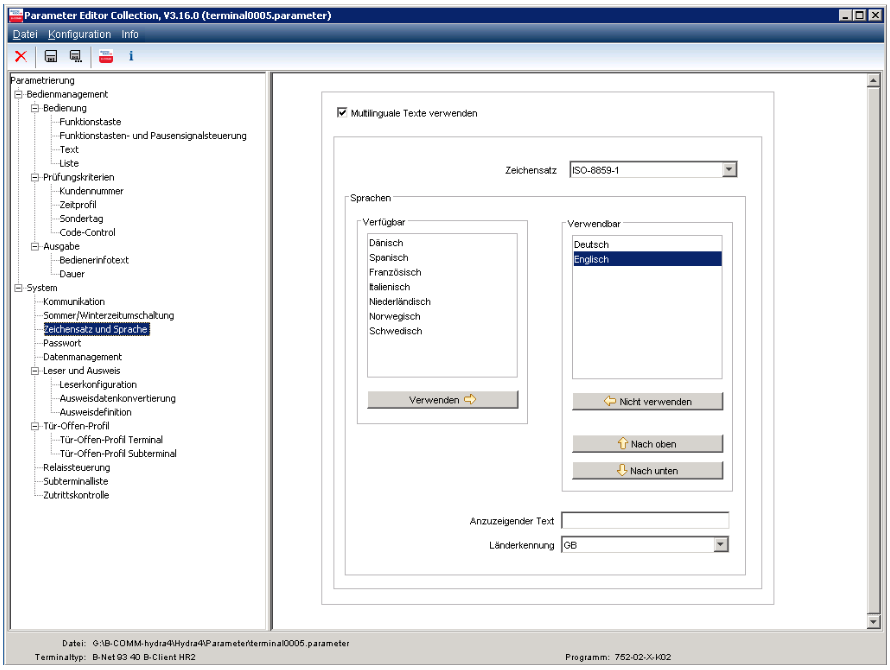
1. Click in the B-COMM menu on "Configuration" and select "Parameter editor".



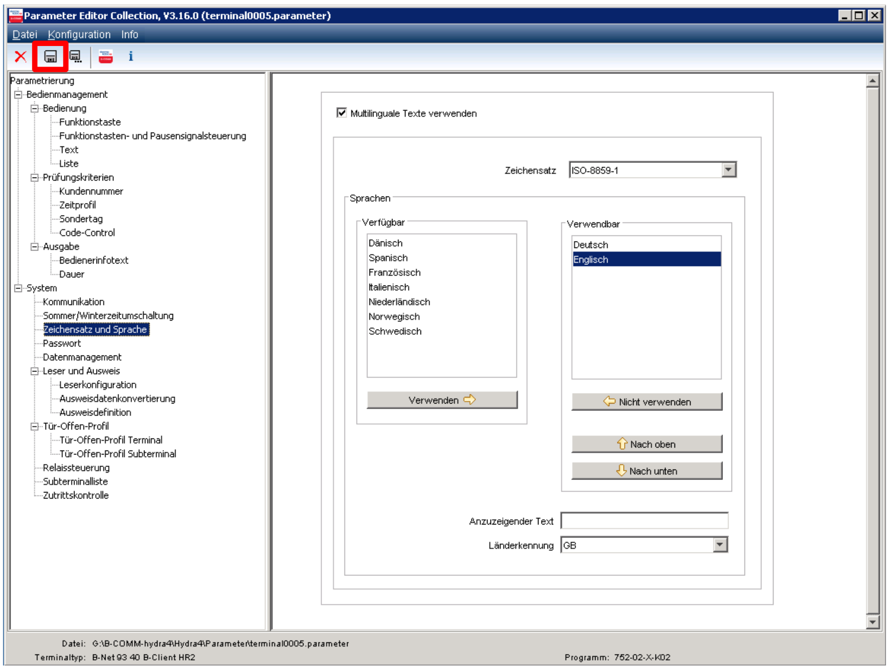
2. Select the KABA terminal that you want to configure and confirm with "Ok".



3. The parameter editor opens. Select in the parameter editor and open in the menu the option "Character set and language".



4. Click on "Use" or "Not use" if you want to select or deselect a language. You can scroll with the button "Upward" or "Downwards" to set the sequence of the KABA hardware language.
5. Press "Save"



6. Reboot the B-COMM after setting the language.

2.5.1.2 Customizing the KABA layout

Refer to the KABA user manual or contact KABA directly if you want to change the KABA hardware layout (e.g. inserting your own company logo).

MPDV does not support customizations of this type. Furthermore, MPDV does not take over any warranty for the effect of a customization of this kind.

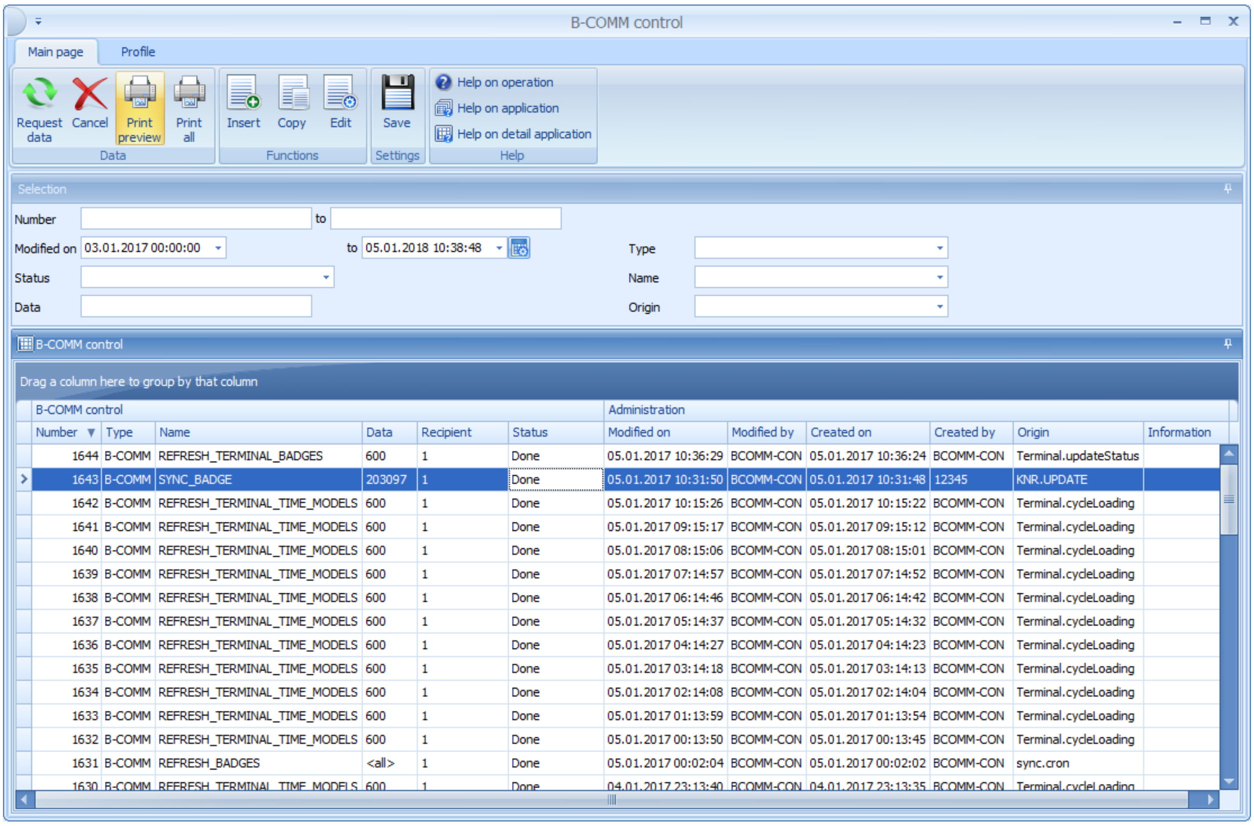
2.5.1.3 Configuration changes

Do not change the following configuration in the B-COMM to ensure the connection from the HYDRA to the KABA hardware functions properly:

- The assignment of the key as the KABA hardware might not be compatible with HYDRA anymore.

3 B-COMM Control

Menu	System administration → Terminals → B-COMM control
Transaction code	bcctrl
Function authorization	bcctrl



Purpose

The application "B-COMM control" is an application of the system administration. Using this application, you can check how the B-COMM connector is triggered by the other functions of the server.

An important feature of the application "B-COMM control" is to identify the badges which were subject to a modification in the MOC and which were - as a consequence - resynchronized with the KABA terminals via the B-COMM connector.

The application monitors how the connector is triggered. Creating or modifying data records is reserved to maintenance or service purposes and is only carried out by MPDV or upon MPDV's instructions.

Requirements

You must install and license the functions required to connect KABA terminals to the B-COMM connector.

Field descriptions

Number

Every time a HYDRA function triggers the B-COMM connector, the system automatically generates a unique consecutive number. The B-COMM connector processes the jobs in the order of these numbers.

Type

Constantly "B-COMM"

Name

Defines the triggered job. Up to now, there is only a single job to be triggered:

UPDATE_BADGE:

This job resynchronizes the badge number entered in the field "Data" with the KABA ZKS terminals.

Data

With name „UPDATE_BADGE“:

This job resynchronizes the badge number entered in the field "Data" with the KABA ZKS terminals.

Status

The following statuses can be entered:

"To do"

The connector is going to process the data record within little time.

"In process"

The B-COMM connector is processing the data record.

"Done"

The data record was processed successfully.

"Done error"

An error occurred during processing. In this case, you can find further information in the error log of the B-COMM connector.

"Cancelled"

You can enter the status "Cancelled" manually for a specific data record. Exceptionally, you assign the status "Cancelled" manually, if the B-COMM connector cannot process the data record for technical reasons.

"Reactivated"

Reserved for future add-ons:

You can assign the status "Reactivated" to a completed data record, if you want to process it again. Technically, this status is identical to the status "To do".

"New"

Reserved for future add-ons:

The data record exists, but cannot be processed yet.

"Unknown" and

"Done unknown"

Reserved for future add-ons.

Priority

A priority is provided for future add-ons. The current priority value is always 50.

Recipient

Specifies the recipient e.g. in case there are several active B-COMM connectors.

Origin

Optional technical ID of the HYDRA function that has triggered the job.

Created by/Created on

Specifies the connector and time of connection.

Modified by / Modified on

Specifies the last connector and the time of modification.

Description of the fields *Name* and *Data*

REFRESH_BADGES

This job updates all badges/master data and the corresponding access authorizations in all terminals. The connector triggers this job on a daily basis right after midnight. As you cannot assign validity periods to the data in KABA terminals, you must update the authorizations in the terminals on a daily basis in case the badges or other configurations expire or are generated in HYDRA.

REFRESH_TERMINAL_TIME_MODELS

The connector triggers this job cyclically. According to the settings in the terminal, the connector reloads the authorizations cyclically. This way, modifications to access time models or opening hours are communicated to the terminals.

The data field provides the terminal number.

SYNC_BADGE

The HYDRA server triggers this job when you modify badges or access profile assignments, which are relevant to the connector. Using this job, you promptly communicate the new badges to the terminals. At the same time, you delete the expired badges in the terminals.

The data field provides the badge number.

The following actions trigger this job:

- You modify the validity status of badges.
- You modify the HR master data and change the validity status of a badge.
- You modify access profile assignments and change authorizations currently valid and relevant to the connected terminals.
- You modify the validity status or the access profile assignment of a badge via the interface HR-PDC (mini HR master data).

REFRESH_TERMINAL_BADGES

This job updates all badges/master data and the corresponding access authorizations in a terminal. The connector triggers this job if it identifies that the option "Reload authorizations" was checked in HYDRA via the terminal administration function.

The data field provides the terminal number.

RELOAD_TERMINAL_BADGES

This job initializes the terminal and updates all badges/master data and the corresponding access authorizations in a terminal. The connector triggers this job if it identifies that the option "Reload program" was checked via the terminal administration function in HYDRA.

The terminal initialization is based on the commands in the terminal classes.

The data field provides the terminal number.

RESTART_TERMINAL

The connector triggers this job if it identifies that the option "Reboot" terminal was checked in HYDRA via the terminal administration function.

NEW_BCOMM_TERMINAL

The HYDRA server triggers this job if you create a new terminal whose terminal type is relevant to the connector. The job is also triggered if the terminal type is modified and hereafter becomes relevant to the connector.

As a consequence, the connector identifies the new terminal and tries to load the data.

Data retention

You can configure the data retention of this application in the Data management for the object *BCOMM-CONN* (product *HYD*). The default value for data retention is 70 days. Once the configured period has expired, the data are deleted.

Installation BCOMM-Connector

Installation

3.1 Requirements

- HYDRA system including Service Pack 12 for HYDRA 8
- Kaba B-COMM installation or installation media for B-COMM including licenses.

3.2 Install HYDRA update

First of all, you install an update of the software for the HYDRA systems where Service Pack 13 (October 2018) has not yet been installed.

The update includes a separate installation guide.

You can find the current system version including service pack information in the file "sp.txt" in the HYDRA directory of the HYDRA server.

In this case, proceed as described in chapter "3.3 HYDRA server: measures"

3.3 HYDRA server: measures

The requirements for the HYDRA server have already been met and no further installation is required if you have a HYDRA system with Service Pack 13 (October 2018) that was installed for the first time.

If this is the case, proceed as described in chapter "0 Check and change **the script file (data management)**

The interface of the B-COMM software uses a new database table to manage actions controlled by time or events. Integrate this table in the data management process to ensure the entries in this database table will be deleted after a specified interval. To do so, manually add an entry to the file *hyarc.scr* in the HYDRA directory of the server.

```
...
# Data management BCOMM control (only with licenses SCS-HCKP or SCS-HCKZ )
if [ `hyliz.exe -r SCS-HCKP` -gt 0 -o `hyliz.exe -r SCS-HCKZ` -gt 0 ]
         then
  echo "Datenmanagement BCOMM-Ansteuerung:" >> $ERRPATH/hyarc.pro
  hymwarc.exe -d -z "PROD=HYD|OBJ=BCOMM-CONN|REORG_VERW_TABLE=1|" > $ERRPATH/hymwarc_hyd_bcomm_conn.pro
  cat $ERRPATH/hymwarc_hyd_bcomm_conn.pro >> $ERRPATH/hyarc.pro

  # File names of system logs
  sleep 1
fi
...
```

HYDRA systems with an initial installation date after September 2017 include service pack 11 and the above mentioned entry.

1. Open the file *hyarc.scr* with a text editor in the HYDRA directory of the server.

2. Use the search function of the text editor to check if the entry already exists. Search for the text *BCOMM-CONN*. If you find the text, skip step 3.
3. Add the entry if it does not exist. Open the file *hyarc.bsp* which includes the entry as a template and copy the entry into *hyarc.scr.*

Import licenses"

If the initial installation of your HYDRA system does not include service pack 13, you need to update the current installation on the HYDRA server. Even if you have installed the current service packs, the system requires an update. This affects HYDRA systems first installed before October 2018.

Among others, this update prepares the HYDRA database for the Kaba B-COMM installation.

Execute database patch

Windows:

Open the folder *HYDRA Administration o*n the HYDRA server desktop and click on the link to start the MS-DOS prompt for the current instance.

Run the following command in the command line:

```
hydscr db_sql/dbp_bcomm_connector.hsc > dbp_bcomm_connector.pro
```

Linux: Connect to the HYDRA server via a telnet connection (for Unix) and start hysys.scr -1 (environment for instance 1).

Run the following command:

```
hydscr.out db_sql/dbp_bcomm_connector.hsc > dbp_bcomm_connector.pro
```

Windows and Linux:

Check if the patch is output in the file dbp_bcomm_connector.pro. No errors or warnings may occur.

This database patch assigns the function authorization for the new application *B-COMM control (bcctrl*) to all users having the function authorization for the application *system logs (syspro).* If necessary, you can still change the assignment of the function authorization *bcctrl*.

Check and change the script file (data management)

The interface of the B-COMM software uses a new database table to manage actions controlled by time or events. Integrate this table in the data management process to ensure the entries in this database table will be deleted after a specified interval. To do so, manually add an entry to the file *hyarc.scr* in the HYDRA directory of the server.

```
...
# Data management BCOMM control (only with licenses SCS-HCKP or SCS-HCKZ )
if [ `hyliz.exe -r SCS-HCKP` -gt 0 -o `hyliz.exe -r SCS-HCKZ` -gt 0 ]
         then
  echo "Datenmanagement BCOMM-Ansteuerung:" >> $ERRPATH/hyarc.pro
  hymwarc.exe -d -z "PROD=HYD|OBJ=BCOMM-CONN|REORG_VERW_TABLE=1|" > $ERRPATH/hymwarc_hyd_bcomm_conn.pro
  cat $ERRPATH/hymwarc_hyd_bcomm_conn.pro >> $ERRPATH/hyarc.pro

  # File names of system logs
  sleep 1
fi
...
```

HYDRA systems with an initial installation date after September 2017 include service pack 11 and the above mentioned entry.

4. Open the file *hyarc.scr* with a text editor in the HYDRA directory of the server.
5. Use the search function of the text editor to check if the entry already exists. Search for the text *BCOMM-CONN*. If you find the text, skip step 3.
6. Add the entry if it does not exist. Open the file *hyarc.bsp* which includes the entry as a template and copy the entry into *hyarc.scr.*

3.4 Import licenses

Import licenses

Make sure that the licenses SCS-HCKZ and/or SCS-HCKP are available. If required, import the licenses via the MOC. Start the application *Licensing* (System administration → System settings → Licensing) and call the function *Insert file*. Select the file *lizenz.dat.*

Restart

After the import of licenses, you must restart HYDRA on the server. The restart must be performed **after** the update has been installed.

If you imported the licenses the day before or at an earlier time, you do not have to restart HYDRA. You do not have to restart HYDRA if you want to proceed with the other steps the next day.

- Quit HYDRA on the server.
  (Use the HYDRA manager for Windows systems, use hy_down.scr for UNIX systems).
- Then restart HYDRA.
  (Use the HYDRA manager for Windows systems, use hy_start.scr for UNIX systems).

3.5 Installation Kaba B-COMM

Follow the B-COMM installation instructions to install the software. Meet the following requirements and recommendations to use the software with the HYDRA-BCOMM connector:

- Kaba recommends the following for the installation:
  Use *Common directory*, but not the program directory. Use a separate directory that includes the program and data. E.g. c:\B-COMM-hydra1.
- Use *bcommservicemanager.bat* to install B-COMM server as a service.
  - The settings for the installation of the service are automatically made.
    - Enter the *Server IP address* and the *rmi control port* as *BCOMM rmi server host* und *BCOMM rmi server port* when installing the BCOMM connector.
    - The other parameters are internal parameters of the B-COMM server. There are no specifications for these parameters, i.e. the internal parameters are not relevant for the connection of the connector.
- Please check if the service is running. If necessary, start the service manually.
- Import license file. Start the B-COMM GUI and select *Import license file* in the menu info/licensing.
- If you have purchased the license for the software option *B-COMM - Option User administration*:

  Log on to the B-COMM GUI as *administrator/111111* for the implementation.
  Kaba recommends:
  If you are prompted to change the password, use the same value 111111 for the *new* password.
  Customers and partners usually do not change this password.
  You should inform MPDV if you change the password. If possible, document the password in the CID.
- Create a new instance, e.g. Hydra1. You also need the name of the instance when installing the connector.
- Select the function New channel in the instance to create a new channel with the following properties:
  - Tab *Network Parameter*
    - Network channel: activated
    - Channel name: BCLAN01
    - Host IP of the server
    - UDP port: specification for the first channel: 30465 = 7701 [hex].

    If you install B-COMM and IL-HYDRA in parallel or if you install several B-COMM instances on one server, make sure to assign unique UDP ports for the communication with the terminals.

    - With IL-HYDRA the port is 7700[hex] + GID, by default. The valid value range is: 7700[hex] to 7729[hex] (30464 to 30505).
    - For B-COMM the valid value range is: 7700[hex] to 77EF[hex] (30464 to 30703).

    If you use IL-HYDRA and B-COMM in parallel, you should start with 7731[hex] (30513) to assign ports in B-COMM. This avoids any overlapping with IL-HYDRA.

  - Tab *Application interface*

    - Do not specify the application interface for the channel first of all. This is only necessary, once the connector has been installed. If the connector is already installed and running, activate the *application interface* and enter the IP address of the HYDRA server.
- Tab *Time synchronization*
    - Activate time synchronization at an interval of 0.5 hours (also for DST/daylight saving time).
- All other tabs: Please do not change default settings.

3.6 Installation of the BCOMM connector

3.6.1.1 Information: HYDRA user *BCOMM-CON*

The BCOMM connector needs a HYDRA user to access the HYDRA services. The system automatically creates this user during the installation using a database patch.

User: *BCOMM-CON*

The HYDRA user is created with a default password. The encrypted password is stored in the connector configuration. The installer encrypts the password during installation. If necessary, you can change the password subsequently in the configuration file of the connector
(e.g. \\SERVER\Hydra\BCOMM_Connector\configuration\application_1.properties)
using the "DB Password Generator".

(If you enter the password with two prefixed hash signs "##" in the configuration files, the system interprets this as an unencrypted password and uses it for the login.)

The user does not need authorizations for the responsibility area.

3.6.1.2 Installation of the BCOMM connector

Start the installation program

<hydradir>\fterm\BCOMM_Connector\setup\BCOMM_Connector\Installer.exe
or installer.sh

Then installation parameters are queried:

Please enter the HYDRA instance

This is the HYDRA system number the connector connects to.

BCOMM connector device id

The first BCOMM connector in a HYDRA installation directory is assigned the number 1. If you want to install several connectors in a HYDRA installation directory, you have to assign unique numbers to the connectors.

BCOMM client name

From Kaba B-COMM: The instance including the channel to be managed (e.g. Hydra1).

BCOMM channel names

From Kaba B-COMM: Name of the channel that should be managed using the connector (e.g. BCLAN01). The indication of several channel is reserved for future use.

Port of BCOMM connector

Unique port that is used to notify the BCOMM connector of changed configurations in HYDRA via html protocol. The port number is e.g. 31101. The third number refers to the HYDRA system number and the fifth/last number to the device ID of the connector.

The connector automatically stores this port number as INI configuration in the HYDRA database. Consequently, the HYDRA root server can trigger the html notification via the web service provider (WSP).

BCOMM rmi server host

Server hosting Kaba B-COMM (server name or IP). The value is assigned during installation of the B-COMM server service or you can find the value in the group "B-COMM server" of the B-COMM starter. Usually, B-COMM is running on the same server as HYDRA.

BCOMM rmi server port

RMI control port of the Kaba B-COMM server (the value is assigned during installation of the B-COMM server service or you can find the value in the group B-COMM server of the B-COMM starter. Usually, the value is 1099).

BCOMM connector application port

Enter this port as port of the *application interface* in the B-COMM channel configuration. The port must be unique in the HYDRA system (Kaba's default value: 3005 decimal).

Address of HYDRA8 WSP

This is the server of the HYDRA system the connector connects to (server name or IP).

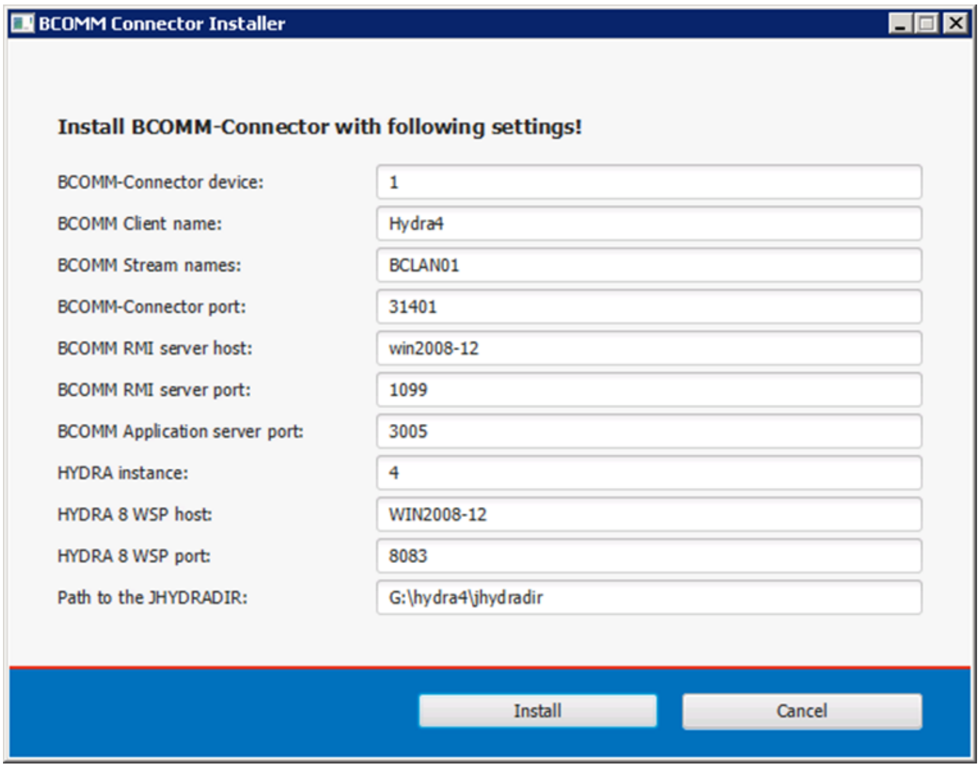
Port of HYDRA8 WSP

This is the port of the HYDRA system the connector connects to.

Path to the JHYDRADIR

Path to JHYDRADIR in the HYDRA installation directory (<hydradir>\jhydradir). This path also defines where the connector is installed. The connector is installed in the directory <hydradir>\BCOMM_Connector.

Once you have entered the installation parameters, a confirmation prompt opens and you can review the settings:



Check the result:

1. A service called HYDRA<n>_BCOMM_Connector_<d> is installed. <n> refers to the HYDRA system number, <d> is the device ID of the connector.
2. The service is started. If the service does not start automatically, start the service manually. The initial automatic start can fail in Windows under certain circumstances.
3. The folder BCOMM\<DeviceId> has been created in the directory <hydradir>\jhydradir.
4. An INI configuration has been created in HYDRA: BCOMMCONNECTOR/ADDRESS/<DeviceId>. The value consists of the HYDRA server and the BCOMM connector port.

Copy terminal classes

Copy from the HYDRA server:

<hydradir>\javadeployment\sample-configs\JHYDRADIR\BCOMM_Connector\1\data

the subdirectory *class* including all other subdirectories into the folder:

<hydradir>\jhydradir\BCOMM_Connector\<DeviceId>\data

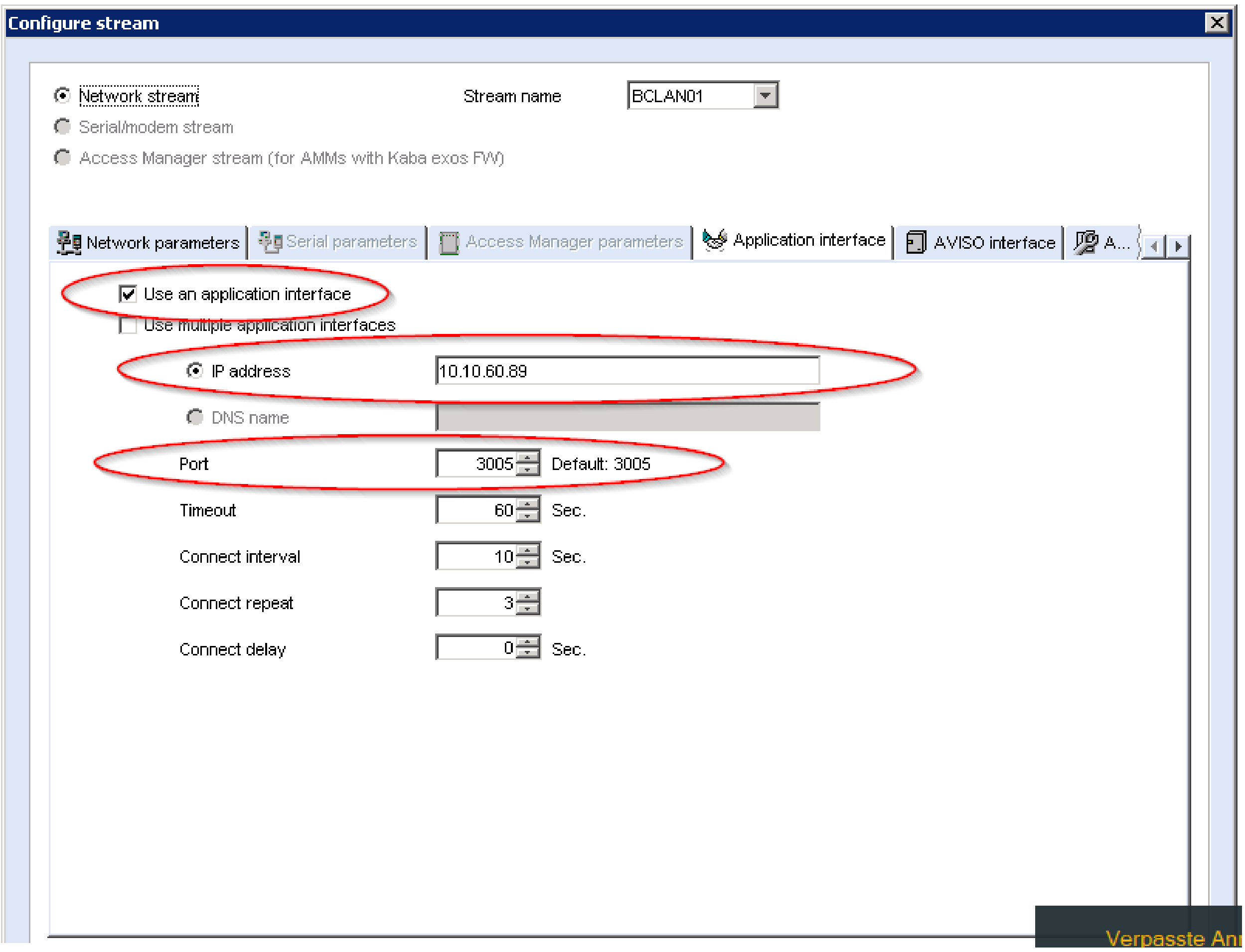
Replace <DeviceId> with the *BCOMM connector device id* assigned during installation (in general = "1"). Overwrite already existing files.

3.6.1.3 Check and change installation parameters subsequently

If required, you can check and change the parameters specified during installation in the configuration file of the BCOMM connector after completing the installation routine. You can find this configuration file in the subdirectory *Configuration* of the BCOMM connector installation directory (e.g. \\SERVER\Hydra\BCOMM_Connector\configuration\application_1.properties).

3.6.1.4 Entry *Application interface* in Kaba B-COMM

Enter the *application interface* in the Kaba B-COMM channel configuration:



Enter the values as described above (previous process steps):

- Configure channel in Kaba B-COMM
  - Tab *Application interface*
    - Activate: *Use an application interface*

- IP or DNS of the server hosting the BCOMM connector (*HYDRA WSP server host*).
- Port (BCOMM connector application port, by default 3005 decimal)
- Other: default settings

Stop the assigned BCOMM connector service to end a channel manually in the Kaba B-COMM.

If the channel is assigned to a BCOMM connector service and the service is running, the service tries to restart its assigned channels at regular intervals.

4 Update

4.1 Install update

Install the MPDV installation medium:

Preparation: Downloading installation medium from the MPDV FTP server

1. Log on to the MPDV FTP server using your access data and download the installation medium for this service pack to your local PC.
2. Save the installation medium downloaded via the FTP server (file extension *.upd)* and other affiliated files in a local directory.

Note: Microsoft Windows Internet Explorer automatically suggests the file extension *zip* when downloading UPD files. For this reason, add the file extension “upd” manually when saving the file. This is not required if you use other web browsers (e.g. Mozilla Firefox).

Perform the following steps for each instance if multiple systems are installed.

Installing the update

The delivery affects the following system components:

HYDRA Server	HYDRA shop floor client	HYDRA MOC	MLE interface
X		X	

1. Start the Maintenance Manager
   Enter the URL *http://<SERVER NAME>:8080/MaintenanceManager* in the web browser
   Enter the IP address or the name of the server where the Tomcat services or processes run in <SERVER NAME>. In general, this is the HYDRA server.
   Note that the URL is case sensitive.

   Use the following URLs to open the Maintenance Manager of the relevant instance if multiple systems are installed:

Instance	Port	URL
1	8080	http://<SERVERNAME>:8080/MaintenanceManager
2	8081	http://<SERVERNAME>:8081/MaintenanceManager
...	...	...

2. Log on with specified password, if required.
3. Select the menu item *Package deployment*
4. Click the *Browse* button to select the *upd* archive (file extension *.upd)* and go to the above-mentioned directory or the data carrier delivered.

5. The button *Deploy package* starts the installation process and shows the log, once the process has been completed.

   Note:
   The installation of the update can take several minutes.

Update the MOC client.

1. Start the MOC client.
2. Go to the menu item *Help* and select *Search for updates.*
3. Click the button *Search for updates.*
4. When the update is found, click *Download updates.*
5. When the updates are downloaded, click *Import updates.*
6. Notes on Microsoft Windows Vista and Windows 7:
   If an MOC client has been installed in the Windows program directory, the Windows user account control (UAC) creates a confirmation prompt. Click *Yes* to confirm.
7. The *MOC Updater* then starts automatically.
8. Click *Start.*
9. When the files have been updated, click *Start MOC* to restart the updated MOC client.

You must update all MOC clients.

4.2 Update the connector

When installing an update with the Maintenance Manager, the BCOMM connector is automatically updated.

The update runs as follows:

- The BCOMM connector automatically identifies the new version. The connector then restarts to activate the current version.

  The new version is copied
  <hydra>\fterm\BCOMM_Connector\setup\BCOMM_Connector\components
  from the directory to the installation directory.

  At the start, the standard *class* files from <hydradir>\javadeployment\sample-configs\JHYDRADIR\BCOMM_Connector\1\data\standard are also copied to the BCOMM connector directory <hydradir>\jhydradir\BCOMM_Connector\<DeviceId>\data.

  (Currently there is no automatic transfer of the *Scope* directories: custom, var and local.)

5 Implementation/configuration of terminals in B-COMM

5.1 Requirements for KABA terminals

5.1.1 General

The terminals must meet specific requirements for the use KABA terminals with the BCOMM connector.

Requirements for KABA terminals for using Time & Attendance (SCS-HCKP):

- The badges must be readable. This might also affect the files *mediaact.ini* and *mediadef.ini*. (B-web 9300, B-web 9600 and B-web 9700).
- The file *Interface.ini* contains the standard interface of a MPDV dormakaba terminal for the terminals of type *B-web 9600* and *B-web 9700.*
- Approved domrmakaba terminals are:

Terminal	Programmversion
B-net 9340 HR2	752-02-X-K02
B-web 9300 HR10	754-00-X-K15
B-web 9600 K5 HR20	736-04-X-K03
B-web 9700 K5 HR30	735-04-X-K03

MPDV provides INI files to implement new terminals or existing terminals. Copy these INI files to the terminal. If you want to keep the settings made on an existing terminal, then you have to compare and synchronize the below-mentioned settings manually.

The INI files provided by MPDV (later also called configuration files) are stored as follows:

HYDRA server →

HYDRA directory →

fterm\BCOMM_Connector\setup\BCOMM_Connector\kaba →

Sub directory (same name as the terminal type e.g.: KABA_HR_97_00_K5)

5.1.2 Configuration files of the terminal

5.1.2.1 Standard configuration files

As a rule, the user copies the INI files provided by MPDV to the terminals. If this is not possible or intended, configure the INI files manually as described below. You cannot use all functions of the B-COMM connection with HYDRA or there might be adverse effects on the terminal behavior if these settings are changed.

In addition to the fixed settings described in the sections that follow, the BCOMM connector sets specific data and settings dynamically depending on how the HYDRA system is configured.

The badge definition must comply with the other reader settings (mediact.ini and mediadef.ini) and does not necessarily correspond to the actual badge number length in HYDRA.

The BCOMM connector sends commands included in terminal classes to the terminal and specifies the settings. The above-described settings refer to the standard terminal classes. Further settings might be set if terminal classes are changed. Refer to the following chapter *Terminal classes* for further information.

5.1.2.2 Required fixed settings for the Time & Attendance (SCS-HCKP)

The fixed settings for a PZE terminal are performed in the B-COMM GUI (e.g. 3.16, B-COMM Server) of dormakaba.

Use the menu item *Configuration => Call parameter editor*... or the context menu of the terminal with *Call parameter editor for assigned parameter file* to request the *Parameter Editor Collection* application where you can fix the settings for a PZE terminal.

You can see the following menu tree in the application:

Parameterization
[-] Operating management
- …

[-] System
- …
- Character set and language
-
- Data management
- Reader and badge

  …
- Defining badge

Execute the following settings in the menu item *Character set and language:*

- [X] use multi lingual text (Default)
- Character set [ISO-8859-1] (Default)

- Language

  You need to configure the languages you want to use on the terminal.

  The first language in the *Can be used* list is the initial language of the terminal after the start.

The languages German, French and English (with the country codes GB and US) are preset in the BCOMM connector.
You need to extend the settings in the BCOMM connector if you want to use an additional language on the terminal.

Store the file here: \\SERVER\Hydra\BCOMM_Connector\configuration\application_1.properties

Execute the following setting in the menu item *Data management:*

- *Master data definition*

  ID lenght [n] should not be bigger or have the same length than the badge number in HYDRA.

Execute the following settings in the menu item *Reader and badge / reader definition:*

- You configure the badge definition for the cards in the menu item *Reader and Badge.* This would be the following configuration for the standard LEGIC badges used by MPDV. Preset

  [user-defined]

	From	Number of digits	ID
	12	05	Customer number
	17	06	Badge number
Other	00	00	Neutral

5.1.2.3 Settings defined by the BCOMM connector (SCS-HCKP)

There are no dynamic settings in the BCOMM connector.

5.1.2.4 Terminal classes

A terminal class is assigned to each Kaba terminal in the HYDRA terminal configuration. Terminal classes define commands for the communication with the terminals.

There is normally no need to change the class files.

You can find terminal classes in the following subdirectory on the server. This subdirectory is part of the HYDRA installation directory.

jhydradir/BCOMM_Connector/1/data/class/standard

5.1.2.4.1 Structure of the terminal class file of Time & Attendance (SCS-HCKP)

A terminal class file is structured like an INI file and consists of three sections:

1) [INIT]
   The section [INIT] includes commands to initialize a terminal as part of the function *Reload program*. The terminal is restarted, once this command has been transferred.
2) [DESIGN] (only for SCS-HCKP)
   There are commandos in the section[DESIGN], which are sent to the terminal when the maintenance function *Reload program* is selected in HYDRA. The standard Icons for the configured *absence reasons* for the terminal B-web 9700 are set here.
3) [REBOOT]
   The section [REBOOT] includes commands that are sent to the terminal if you select the HYDRA maintenance function *Reboot*.
4) [AFTER_INIT]
   The section [AFTER_INIT] includes optional commands that are sent to the terminal after initialization. The section [AFTER_INIT] only allows commands that do not require a restart. By default, this section is empty.

The individual commands are structured as key pairs.

Example

```
setOperatingMode = "@T T1"
```

Key: `setOperatingMode`

Command: `"@T T1"`

Enclose the command in double quotes.

The first two characters of a command stand for the GID/DID. This GID/DID is replaced with the terminal's GID/DID when transferring the command to a terminal.

Comments start with a semicolon.

5.1.2.4.2 Replacement of values in class file commands

The commands included in the class file are transferred to the terminal in order to initialize the terminal. Some commands contain specific values/fields that are replaced with contents from the HYDRA configuration.

5.1.2.4.3 General replacements

The GID/DID of all commands is replaced with the GID/DID of the target terminal.

5.1.2.4.4 Constraints

You can control the replacement of values in commands. To this end, you can define constraints for each command. Constraints are mainly required to customize class files (see following chapter).

Insert a space character to add constraints to the command that is enclosed by double quotes, e.g.

```
initSpecialDay2 = "@T >3L0200000000000000000000000000000000000000000000000000" NoDataReplacement
```

NoDataReplacement

The constraint *NoDataReplacement* only replaces the *GID/DID* and no other data fields, even if they are described above.

RAW

The constraint *RAW* does not replace data, not even the *GID/DID*. The command is only sent to the terminal matching the entered GID/DID.

GIDDIDSubtermXX

Use the constraint *GIDDIDSubtermXX* to replace the *GID/DID* with the GID/DID of the sub-terminal. XX stands for the consecutive reader number. Thereby, you can send commands directly to a sub-terminal. (At the moment there is no reasonable application for this option in the standard).

5.1.2.4.5 Customization of a class file

Usually, you do not have to customize terminal classes. If you customize the files, you have to observe the scope concept.

The terminal class directory includes several subdirectories. The class files delivered by MPDV are stored in the standard directory. You can file customizations to these standard class files in the other directories.

The system first reads the standard file and saves the commands per section and key in the specified order. Then the system reads the class files of the other scopes in a defined order and executes the specified commands.

- If the section already includes a command for the key, the command is overwritten. If you assign an empty command, the command is deleted.
- If the section does not yet include the key, the key is added to the existing commands in this section.

Priority	Directory	Contents
1 (lowest)	standard	Class files delivered by MPDV. You must not customize the files in this scope. MPDV updates can overwrite these class files at any time.
2	custom	Optional customizations carried out by MPDV. MPDV updates can also overwrite these files at any time.
3	var	Optional customizations carried out by a partner or value added reseller (VAR). These files are not overwritten by MPDV, but can be modified by the partner/reseller.
4 (highest)	local	Optional customizations carried out by the end customer. These files are not overwritten by MPDV and should not be modified by a partner or reseller. Only end customers should modify these files.

5.1.2.5 The terminal class file of Time & Attendance (SCS-HCKP)

The following settings/configurations are transferred from the BCOMM connector to the dormakaba terminal during initialization by the *Reload program*. The data from the terminal-specific class file (e.g. class90.ini for *B-web 9700)* is used. Whereby parts of the configured command are replaced with data from HYDRA.

The languages German, French and enEnglish (GB) and enUS(English US) are preset in the BCOMM connector.

The configurations described in this chapter can only be changed after consultation with the MPDV.

5.1.2.5.1 General text >T01-28 and >t01-28 (multi)

The general texts are transferred to all PZE terminals. The texts are displayed on the dormakaba terminal after an action as employee information (e.g. *Please wait, Read error).*

5.1.2.5.2 Dialog texte >D00-15 and >d00-15 (multi)

The dialog text are transferred to all PZE terminals. The dialog text are shown if you click on a certain field (e.g. *Badge please)*

5.1.2.5.3 Funktion keys texte >M00-05,31-40 and >m00-05,31-40 (multi)

All function key text are transferred to the PZE terminals. These are copied from the terminal label => tab *HR functions* of HYDRA.

5.1.2.5.4 Funktion keys texte >F00-05,31-40 and >m00 (multi)

The initialization of the data takes place after the configuration in the terminal label→ Tab HYDRA HR functions

5.1.2.5.5 Parameter sets ">3z00"

With the entry: "@T >3z00init/interface.ini[SurfaceDesign]DisplayInfoVisibleNumber=10" the number of lines to be displayed in the info is set to 10. (This setting is only required in the file *class85.ini* for a dormakaba terminal B-web 9600)

5.1.2.5.6 Other configurationen >3E01, >3X02 , >3X01

With the entry: *@T >3E01----03----* the *operating language* and the *seconds* in the data records are activated in the dormakaba terminals.

With the entry: *@T >3X02------------XXXXXXXXXX------------* the system company number to be checked is transferred to the dormakaba terminals. The placeholder XXXXXXXXXX is replaced by the data from the terminal label (or HYDRA setup).

A "-" digit is not checked.

With the entry: "@T >3X0105000500050003001200000" the times for the terminal control are transferred to the dormakaba terminals.

Data derives from the following fields in the *terminal label → Tab HYDRA HR functions*.

dormakaba terminal time	**HYDRA data field**
Display time *Authorized*	Return time
Display time *Not authorized*	Return time
Pickup time relays	Relay time
Operating timeout	Return time
Display time disply info	Display duration of info

5.2 Configure terminal in Kaba B-COMM

Kaba supports the following values for GID/DID: GID: 00 to 29, DID: 00 to 59.

If you use the communication software B-COMM, you have to consider the following when you assign the device ID *DID:*

5.2.1 Terminal to collect Time & Attendance (SCS HCKP)

You cannot use sub terminals in combination with PZE terminals.

Procedure:

- Configure new network adapter
  - Enter the terminal's IP address
  - Enter GID/DID
- Log on terminal
  - Set operating status online
  - Other time zones are currently not supported

5.3 Configure terminal in HYDRA

You configure the terminals in HYDRA with the following path:

System administration → Terminals → Terminal configuration

The terminal must be configured in HYDRA. Choose a short *Cycle duration of status messages* at the beginning. Therefore, terminals are quickly initialized and provided with authorizations.

Then, trigger the function *Reload program* via the terminal administration.

You can check completion of the *Reload program* process in the MOC application *B-COMM control*.

5.3.1 Configuring the terminal for Time & Attendance in HYDRA

For information on how to configure the terminal for Personnel Time Management, See here.

6 Check result and troubleshooting

6.1 Logging

6.1.1 Status page of the BCOMM connector

Use a web browser to start a status page of the BCOMM connector:

To do so, enter the address of the BCOMM connector in the address line. You can find the address in the *INI data configuration* of the MOC:

Name	BCOMMCONNECTOR
Section	ADDRESS
Key	Instance number, e.g. *1*
Value	Address of the BCOMM connector, e. g. *servername:31101*

Tabelle 1.1

Prefix http:// to the value from the *INI data configuration*, attach */status* and enter this as address in your browser, e.g.:

http://servername:31101/status

A status page is shown:


```
{
  "version": "1.0.3469",
  "startupDate": "2018-06-01T12:29:08.877+02:00",
  "uptime": "D:2 H:20 M:35 S:6",
  "totalMemory": 262668288,
  "freeMemory": 141175448,
  "workingDir": "file:/G:/hydra4/BCOMM_Connector/BCOMMConnector-1.0.3469.jar!/BOOT-INF/classes!/",
  "osArch": "amd64",
  "osName": "Windows Server 2008 R2",
  "osVersion": "6.1",
  "cpuLoad": 8.3,
  "registeredBcommchannels": 1,
  "notifiedFromHydra": true,
  "dataTransferWithHydra": true,
  "initializedSuccessful": true,
  "queueSize": 0,
  "clockingQueueSize": 0,
  "terminalInfo": [
    {
      "terminalId": 704,
      "terminalType": "PZE",
      "terminalClass": 85,
      "ipAddress": "10.10.11.46",
      "gid": 1,
      "did": 2,
      "authorizationInfos": {
        "mirror": {
          "size": 36,
          "time": "2018-06-04 09:03:07"
        }
      }
    }, ..
        ]
}
```

You can install a plug-in for your browser to display JSON if your browser output is not as easy to read as shown above. You can find such plug-ins for common browsers on the internet.

Meaning of data

Apart from the general, self-explanatory values like *version, uptime* or *workingDir,* there are also special values for the BCOMM connector status:

registeredBcommchannels

Number of Kaba B-COMM channels logged at the connector. As soon as channels are logged on to the BCOMM connector (i.e. a connection from BCOMM to the connector is possible), here the number of channels connected to is counted up. If the number of logged channels is greater than 0, you can exchange data between the BCOMM connector and the Kaba-B-COMM.

The Kaba B-COMM logs the channel with the BCOMM connector if the channel is restarted.

The BCOMM connector restarts the channels assigned to the Kaba B-COMM if the BCOMM connector is restarted. The system also checks every minute if the registered channels are still active. Inactive channels are also restarted.

Stop the assigned BCOMM connector service to end a channel manually in the Kaba B-COMM. If the service is running, the service tries to restart its assigned channels at regular intervals.

Proceed as follows in case of problems:

Check if the correct application interface is defined in the Kaba B-COMM.

Check the settings of the Kaba B-COMM and the channel according to the BCOMM connector's installation guide.

The *server IP address* and the *rmi control port* of the Kaba B-COMM must match the BCOMM connector's *BCOMM rmi server host* and *BCOMM rmi server port.*

notifiedFromHydra

For specific events, the HYDRA server notifies the BCOMM connector. Consequently, the BCOMM connector can immediately execute *B-Comm control* activities. This is the case, if a new badge is created or a terminal assigned to the BCOMM connector is changed. The variable switches to *true*, once the HYDRA server has informed the BCOMM connector.

Proceed as follows in case of problems:

A firewall might prevent the HYDRA server from reaching the BCOMM connector via the address defined in the *INI data configuration*.

The address stored in the *INI data configuration* might not be correct (see table 1.1). This address is normally entered by the BCOMM connector at the startup if the HYDRA server can be reached successfully by the BCOMM connector.

dataTransferWithHydra

true: Communication of the BCOMM connector with HYDRA via services has taken place at least once. HYDRA can read and send the data.

Proceed as follows in case of problems:

Check the HYDRA server settings and compare these settings with the one's in the BCOMM connector's installation guide. Also check the communication user BCOMM-CON. Refer to chapter 4.5.3 for further information on the check.

6.1.2 Application: B-COMM Control

Menu	System administration → Terminals → B-COMM control
Transaction code	bcctrl
Function authorization	bcctrl

Use the B-COMM control application to trace specific interface activities. The columns *status* and *information* provide information on the status of each activity. You can find application details in the associated documentation and online help.

6.1.3 HYDRA system logs

The HYDRA system logs include an entry for the application BCOMMCON called *B-COMM Connector 1 error log*. This log records any critical issues of the connector.

6.1.4 Log files of the BCOMM connector

The connector is generally logged in `<hydradir>/BCOMM_Connector/log`. By default, this directory logs the levels *error* and *warnings/alerts*.

If required, you can increase the log level. To do so, open the file `application_X.properties` (X stands for the device number of the connector) in the directory `<hydradir>/BCOMM_Connector/configuration/` and change the entry `logging.level.de.mpdv=WARN`.

Possible log levels:

- ERROR error
- WARN unexpected situations occur
- INFO general information

- DEBUG debug messages
- TRACE detailed debugging messages

Please note that the lower log levels (trace) also include the higher log levels (ERROR, WARN, ...).

Restart the connector after changing the log level.

The entry `logging.level.de.mpdv` affects all MPDV log messages. You can also change the log level for individual classes. In this case, you should know the package/class name. If you only want to increase the log level for translating BCOMM messages, create another entry for the translation package *babelfish*:

```
logging.level.de.mpdv.bcomm.babelfish=INFO
```

6.2 Mirror files

The BCOMM connector stores the current status of terminal data in mirror files. Mirror files are stored in the sub-directory `mirrors` of the directory `<hydradir>/jhydradir/BCOMM_Connector/<DeviceNr>/data`.

There is one JSON file for each terminal. The connector keeps the files and uses the files as cache of terminal data and authorizations.

Start *reload program* in the terminal configuration to restructure the mirror file.

If you delete the file manually, you must perform the option *reload program* in the terminal configuration in order to ensure the mirror files will be restructured. Otherwise, the next data synchronization might result in wrong time authorizations and missing or excess badges.

6.3 Terminal classes

Changes to terminal classes could lead to malfunctions. If you suspect this, check the terminal classes and compare these classes with the factory settings. (See chapter 3.1.2.4)

6.4 Upload a parameter file of the terminal

For diagnostic purposes, you might require a parameter file upload from the Kaba terminal. You can upload the file via the Kaba B-COMM GUI.

Always request the current version of the parameter file from the terminal!

1. In the Kaba B-COMM GUI open the context menu with a right click on the required terminal and choose the option *Upload/assign parameter file*. If necessary, confirm any message that appears about overwriting an existing file.

2. Check if the upload was successful: the channel should include ParameterUpload with status FinishedOK as the last job.
3. Check data: use the parameter editor to view the parameter file.
4. Storage location of the parameter file: if you need the parameter file for analysis/backup purposes, you can identify the storage location of the files via the *Parameter file assignment* in the menu *Configuratio*n.

6.5 Possible issues

6.5.1 Invalid badges

You have created a new badge in HYDRA but this badge is not yet valid for the terminal.

Possible reasons and solutions:

1. B-COMM control does not work:
    a. Check if the synchronization job SYNC_BADGE of the application B-COMM control has been processed.
    b. Are there any other jobs scheduled before?
    c. Is there a job that got stuck?
    d. If there is no other job, you can still use the options *Reload authorizations* or *Reload program* from the terminal configuration. These options synchronize the data. The option *Reload authorizations* only transfers the changed data. The option *Reload program* reinitializes all terminal data.
    e. Check the status page of the B-COMM connecter to identify the cause of the problem.

2. An access attempt is rejected with the message *Unauthorized badge* in the access log.
    a. The access authorizations in HYDRA show that the badge has no authorizations at this time.
    b. The badge has not yet been synchronized (see issue a)).
    c. Check the status page of the B-COMM connecter to identify the cause of the problem.

6.5.2 New terminals do not receive data

A new terminal was created in HYDRA, but no data is transferred to this terminal.

Perhaps, the connector has not yet received any information on this terminal from HYDRA or the BCOMM. Restart the B-COMM connector to solve the problem.

Check installation of the Kaba terminal. The used INI files might not comply with the requirements for the B-COMM connector. A cold start of the terminal can solve the problem. If necessary, copy the INI files provided by MPDV onto the terminal. See chapter "5 Implementation/configuration of terminals in B-COMM“.

6.5.3 Check communication user BCOMM connector

Error pattern

It looks like the BCOMM connector cannot read any data from HYDRA. Entries of the B-COMM control are not processed and newly created badges, accesses or terminals are not identified. Nothing works after having rebooted the BCOMM connectors.

There are entries in the log file of the BCOMM connector (normally in the directory <hydradir>/BCOMM_Connector/log) that indicate errors during user logon.

Info

The BCOMM connector needs a HYDRA user to access the HYDRA services. The system automatically creates this user during the installation using a database patch.

User: *BCOMM-CON*

Create the HYDRA user with a default password. The encrypted password is stored in the HYDRA database and in the connector configuration. The installer encrypts the password during installation.

If necessary, the password can subsequently be changed in the HYDRA user administration and in the configuration file of the connector (e.g. \\SERVER\Hydra\BCOMM_Connector\configuration\application_1.properties) with the *DB password generator*.

(If you enter the password with two prefixed hash signs "##" in the configuration files, the system interprets this as an unencrypted password and uses it for the login.)

The user does not need authorizations for the responsibility area.

Solution

The configuration of the communication user may be faulty:

- Perhaps the user was deleted by mistake or not created during the installation.
- The passwords of the user in HYDRA (unencrypted input at the MOC) and in the configuration file (encrypted input) may not match.

There are several solutions:

1) Restore default password for existing user
    a. The column pwd must have the content '324b8eff4d96b147c761acec08537d06'. In the database table user_tab for the record with usr = 'BCOMM-CON' Check the value of the column pwd and, if necessary, restore it using SQL.

    b. Check the following entry in the connector's configuration file and restore it if necessary: mpdv.mw.password=324b8eff4d96b147c761acec08537d06

2) Create the HYDRA user with a default password again:
    a. Delete the user BCOMM-CON on the MOC if there.
    b. Execute the above described patch dbp_bcomm_connector.hsc again. Create the patch with default values.
    c. Check the following entry in the connector's configuration file and restore it if necessary: mpdv.mw.password=324b8eff4d96b147c761acec08537d06
3) Check and correct the user with a separate password.
    a. Create the user BCOMM-CON on the MOC if not there. Remember the assigned password.
    b. If the user exists and the password is unknown, assign a new password and remember the password.
    c. Encrypt the new password with the *DB password generator.*
    d. Enter the encrypted password in the configuration file of the connector in the key mpdv.mw.password=.